VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi-590018

A Project Report on
"Smart Assignment and Feedback Management System"

Submitted in Partial Fulfillment of the Requirements for the IV Semester of the Degree of

Bachelor of Engineering in Computer Science & Engineering

By

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Under the Guidance of

Dr. Prem Kumar Ramesh, Associate Professor, Dept. of CSE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CMR INSTITUTE OF TECHNOLOGY

Affiliated to VTU, Approved by AICTE,

Accredited by NBA and NAAC with "A++" Grade

ITPL MAIN ROAD, BROOKFIELD, BENGALURU-560037, KARNATAKA, INDIA

Academic Year: 2025–2026

CERTIFICATE

This is to certify that the Mini Project work entitled **"Smart Assignment and Feedback Management System"** has been carried out by **Zeba Fatma (1CR23CS220)**, a bonafide student of CMR Institute of Technology, Bengaluru in partial fulfillment for the award of the Degree of Bachelor of Engineering in Computer Science and Engineering of the Visvesvaraya Technological University, Belagavi during the year 2025–2026.

This mini-project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said Degree.

Dr. Prem Kumar Ramesh	**Dr. Kesavamoorthy R.**
Associate Professor	Professor & Head of Department
Dept. of CSE, CMRIT	Dept. of CSE, CMRIT

DECLARATION

I, **Zeba Fatma (1CR23CS220)**, student of the V Semester, Department of Computer Science and Engineering, CMR Institute of Technology, hereby declares that the project entitled **"Smart Assignment and Feedback Management System"** has been successfully completed under the guidance of **Dr. Prem Kumar Ramesh**, Associate Professor. This report is submitted in partial fulfillment of the requirements for the award of the Degree of Bachelor of Engineering in Computer Science and Engineering for the academic year 2025–2026.

The contents of this report have not been submitted to any other university for the award of any degree or diploma.

Place: Bangalore Date: November 26, 2026

Zeba Fatma (1CR23CS220)

ABSTRACT

The Smart Assignment, Testing Feedback Management System (SAFMS) is a full-stack web application that digitizes academic workflows such as assignment management, online examinations, grading, and real-time performance tracking. The platform enables teachers to create assignments and initiate MCQ-based tests with specified durations. Students can participate in live tests strictly within the test window, and scores are calculated automatically in real time. A dynamic leaderboard displays rankings instantly based on student performance. The system includes secure authentication, file-handling capabilities, REST API architecture, and a responsive frontend. SAFMS enhances efficiency, transparency, and accessibility in academic environments, making it suitable for online learning use cases.

ACKNOWLEDGEMENT

I take this opportunity to express my sincere gratitude to CMR Institute of Technology, Bengaluru for providing me with the opportunity to carry out this Project.

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I thank **Dr. Kesavamoorthy R.**, HOD, Department of CSE, for his guidance.

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I am also thankful to all faculty members and my peers for their encouragement throughout this project.

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1. INTRODUCTION

In academic institutions, managing assignments and conducting assessments manually is time-consuming and inefficient. The Smart Assignment, Testing Feedback Management System (SAFMS) solves these challenges by offering a unified digital platform for teachers and students. This system supports:

- Creating and managing assignments
- Conducting MCQ-based online tests
- Enforcing time-bound participation
- Real-time test scoring
- Leaderboard generation
- Submission review and feedback

SAFMS consists of a React frontend communicating with a Node.js + Express backend using REST APIs. The backend supports file uploads, JWT authentication, and either file-based or MongoDB storage. The design promotes modularity, scalability, and easy enhancement for future academic features.

2. SOFTWARE REQUIREMENTS

Programming Languages:

- JavaScript (Node.js, React)

Frontend:

- React.js

Backend:

- Express.js

Database:

- JSON file storage (default) or MongoDB (optional)

Tools Libraries:

- Node.js v16+
- npm
- multer (file uploads)
- JSON Web Tokens (JWT)
- axios
- cors

3. Implementation

3.1 System Architecture

3.1.1 Frontend (React.js)

The frontend of the system is developed using React.js and includes the following components:

- Authentication interface
- Assignment dashboard
- Test interface with MCQ rendering
- Real-time score updates
- Live leaderboard display

3.1.2 Backend (Express.js)

The backend is built using Express.js and provides the core functionality through REST APIs:

- Auth routes (login/register)
- Assignment management
- Test management (create, start, end)
- MCQ evaluation logic
- Real-time scoring and leaderboard
- Submission and file management

3.1.3 Storage Options

The system supports two different storage mechanisms:

- **Default:** Local `db.json` file-based storage
- **Optional:** MongoDB database using Mongoose models

3.2 Major Functional Modules

3.2.1 A. Authentication Module

The authentication module ensures secure access and role-based control:

- Supports Teacher and Student roles
- JWT-based secure session handling
- Role-protected backend routes

3.2.2 B. Assignment Module

This module handles assignment creation and submissions:

- Teachers create assignments with title, description, and deadline
- Optional question PDFs and marking scheme PDFs can be uploaded
- Students can view assignments and submit their responses

3.2.3 C. MCQ Test Module (Updated Feature)

This module enables fully online, time-bound MCQ-based testing.

Teacher Functionality

- Create MCQ-based tests
- Add multiple questions with options
- Set test start-time and end-time
- Start the test
- Monitor live student participation

Student Functionality

- Students can join the test only within the active duration
- Test automatically locks after the deadline
- Answers are submitted automatically

System Functionality

- Automatically calculates scores based on correct MCQs
- Stores all responses securely

3.2.4 D. Real-Time Scoring System

The scoring system evaluates answers and updates scores immediately:

- Each submitted answer is evaluated instantly
- Total score updates dynamically
- Efficient for large sets of questions

3.2.5 E. Live Leaderboard Module

The leaderboard system ranks students based on performance:

- Displays ranking of all participating students
- Updates automatically as scores change
- Visible only after the test starts
- Helps track top performers in real time

3.2.6 F. Doubt Management

This module enables academic discussion and clarification:

- Students can post doubts
- Teachers can reply to doubts
- User identity is always visible
- Anonymous posting is not permitted

3.2.7 G. Student Analytics Module

The Student Analytics module provides a comprehensive overview of a student's academic performance across assignments and tests. It is designed to help students track their progress, identify strengths, and understand areas that require improvement.

- Displays all submitted assignments along with grades and teacher feedback.
- Shows detailed test scores for every MCQ-based test attempted by the student.
- Provides visual charts and graphs summarizing performance trends over time.
- Highlights leaderboard positions in completed tests.
- Includes overall statistics such as average score, highest score, and completion rate.
- Presents attendance in tests and submission consistency.
- Offers a unified summary dashboard for a clear academic overview.

This module enhances student self-evaluation by presenting organized, data-driven insights into their academic progress.

4. Result – Output

The Smart Assignment, Testing & Feedback Management System (SAFMS) successfully delivers a complete digital workflow for assignment management, MCQ test conduction, real-time scoring, leaderboard updates, doubt resolution, and student performance analytics. This section presents the final working outputs and interface screenshots.

Assignment Management

- Teachers can create assignments with questions, PDFs, and deadlines.
- Students can view assignments, upload submissions, and receive evaluated scores.

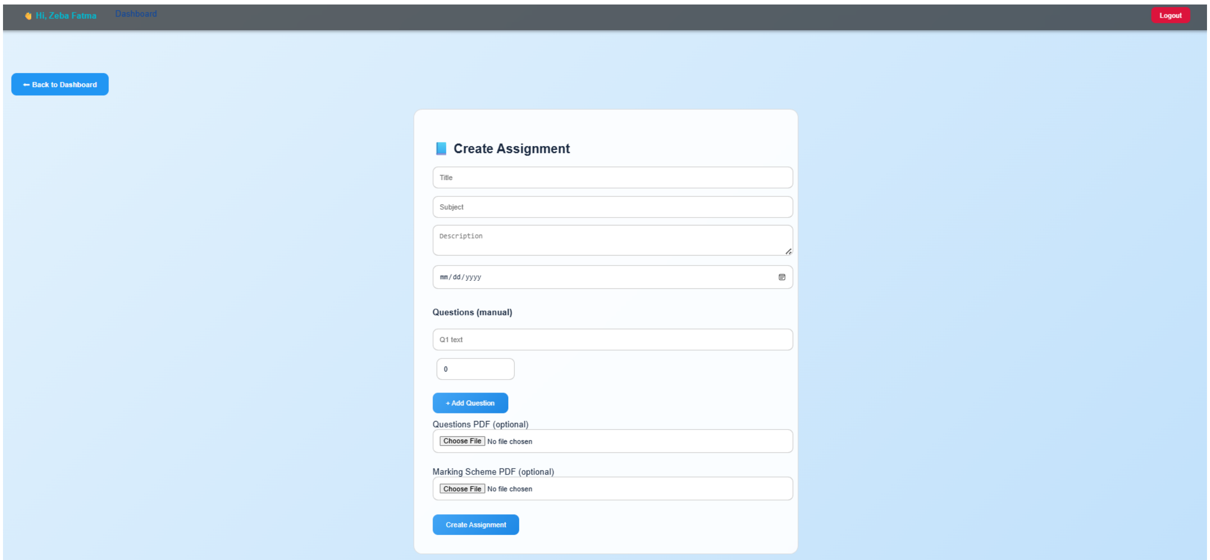

Figure 1: Teacher – Create Assignment Interface

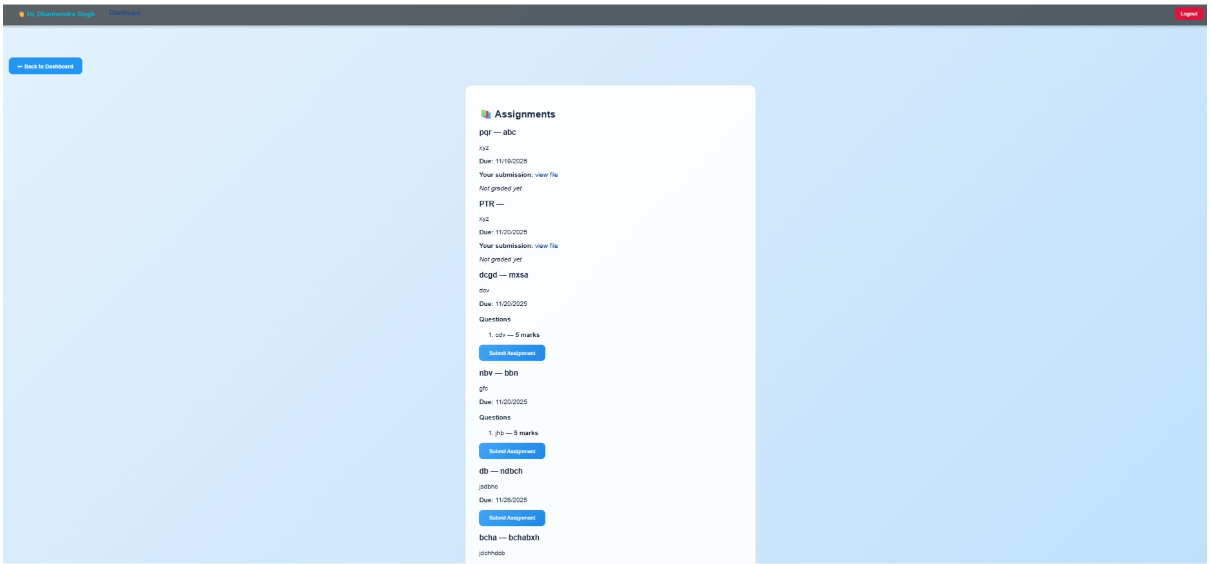

Figure 2: Student – View Assignments Interface

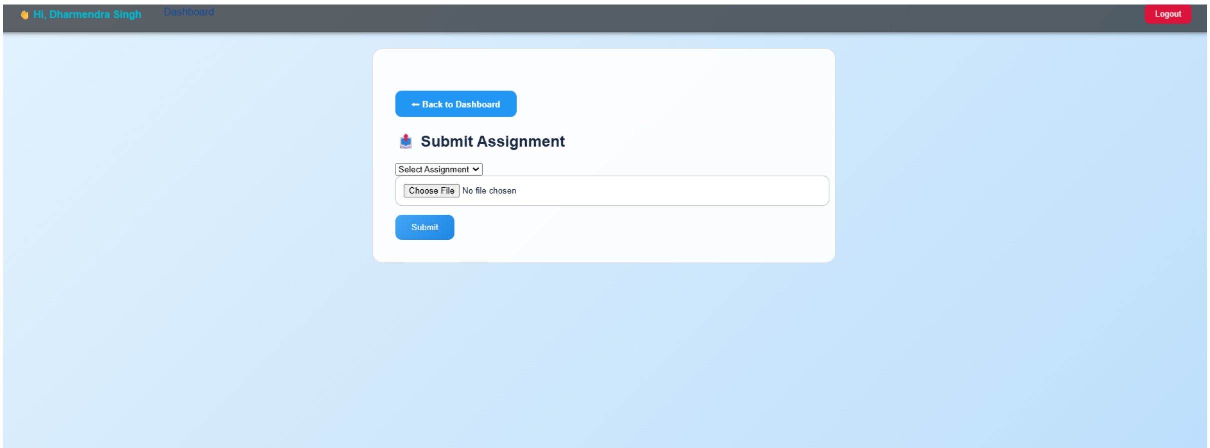

Figure 3: Student – Submit Assignment Page

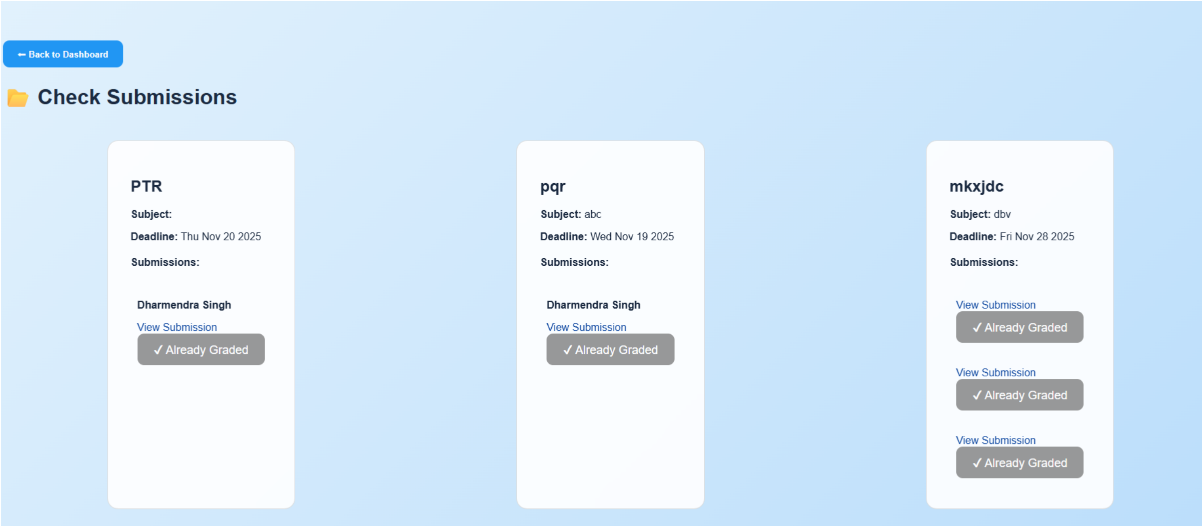

Figure 4: Teacher – Check Submissions and Grade

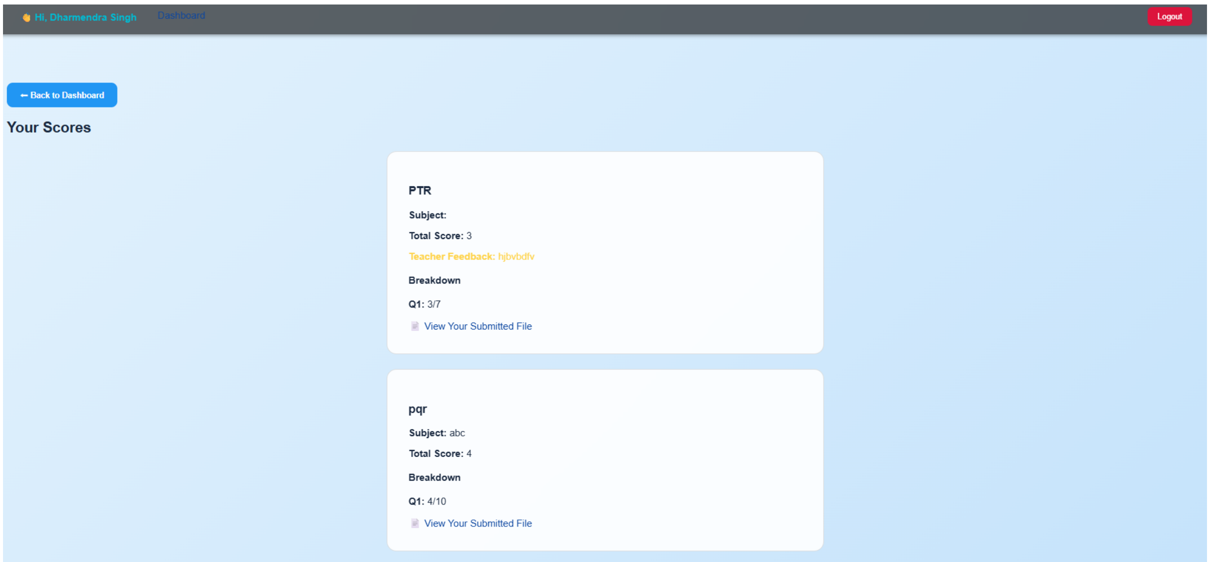

Figure 5: Student – View Assignment Scores and Feedback

MCQ Test Management

- Teachers create MCQ tests with duration and questions.
- Students can attempt tests only during the active test window.
- Tests auto-submit and are scored instantly.

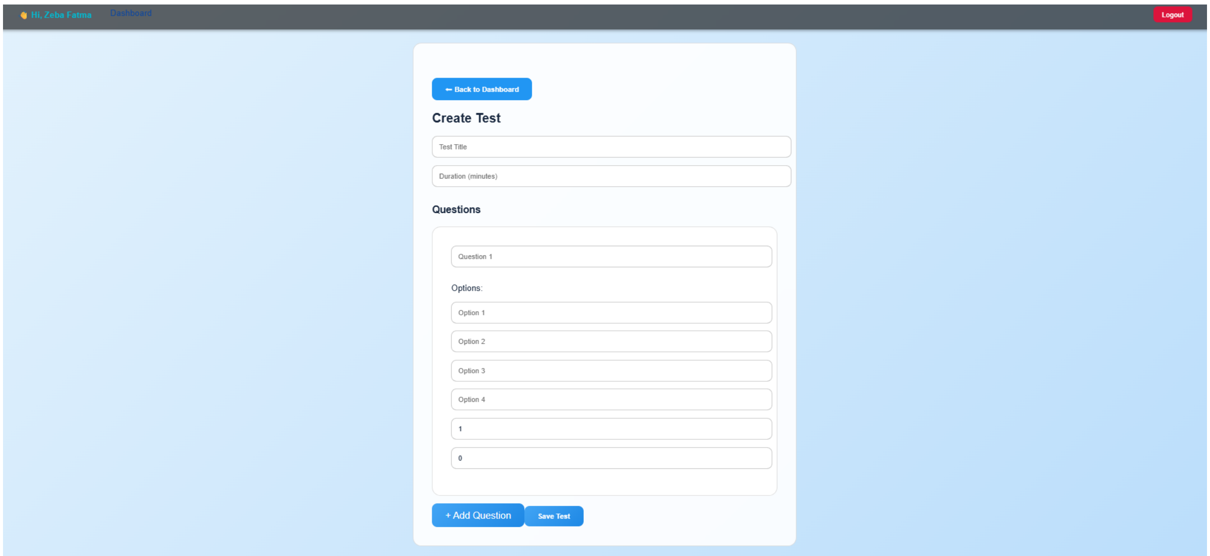

Figure 6: Teacher – Create MCQ Test Interface

Figure 7: Student – Active Test Attempt Screen

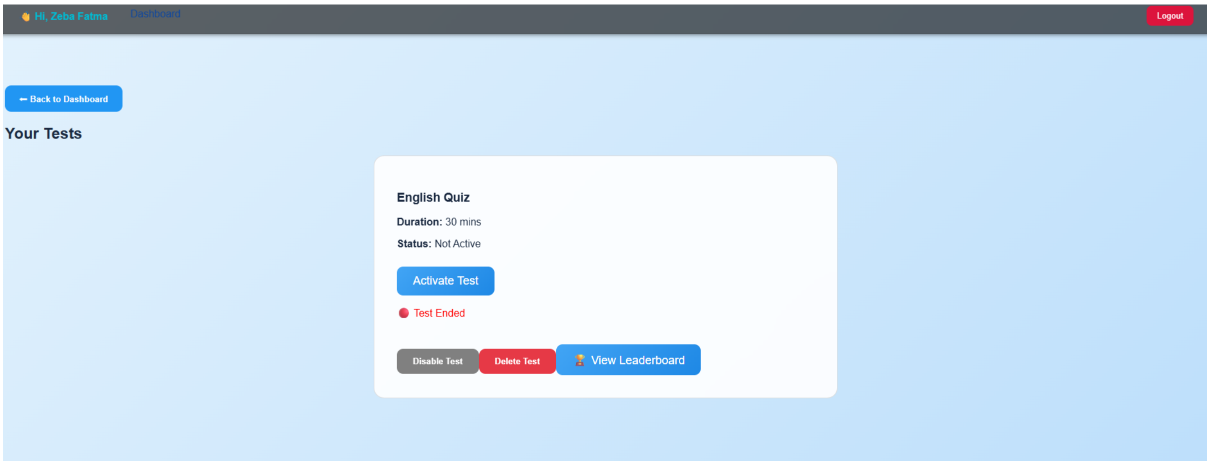

Figure 8: Teacher – Test Management and Activation Panel

Real-Time Scoring and Leaderboard

- MCQ answers are auto-evaluated immediately.
- Scores are updated in real time.
- Leaderboard displays ranks dynamically.

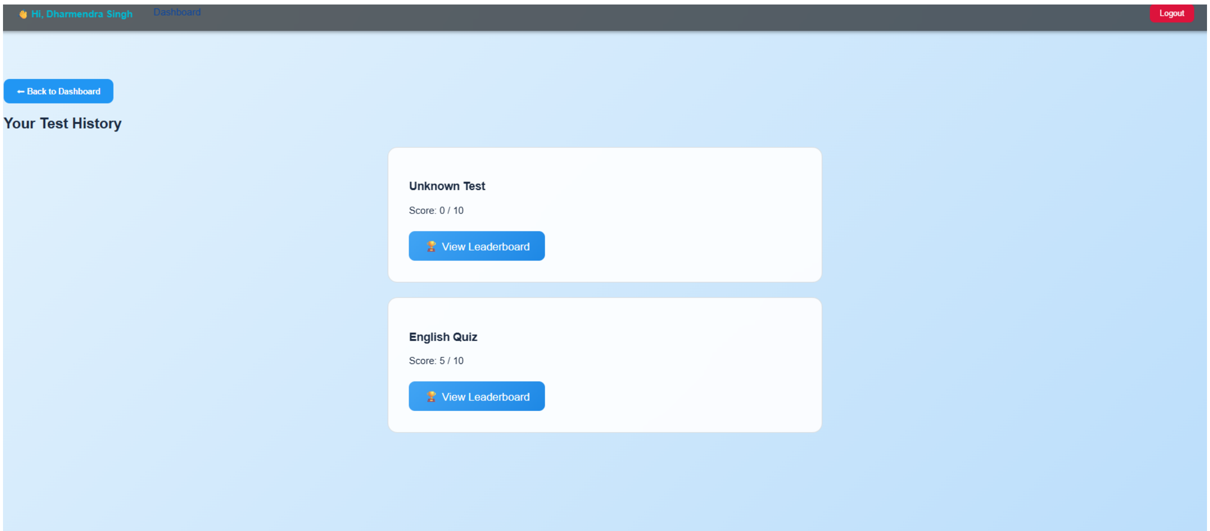

Figure 9: Student – Test History with Scores

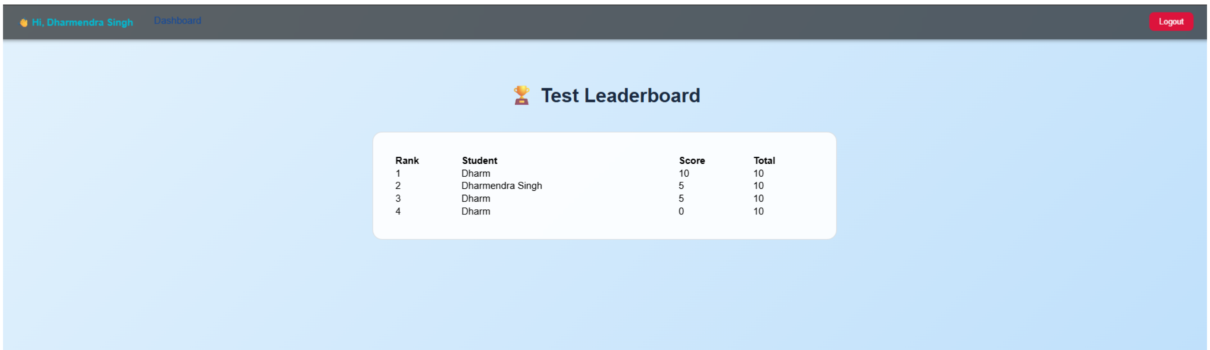

Figure 10: Student – Test Leaderboard View

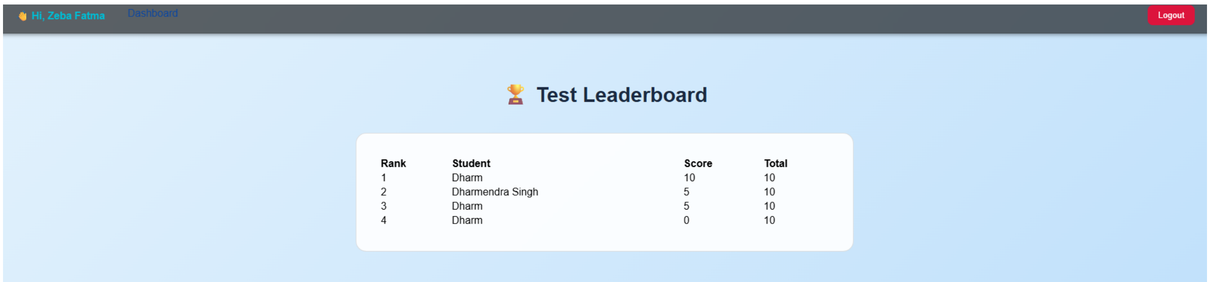

Figure 11: Teacher – Leaderboard Overview

Doubts and Problem Solving

- Students post doubts publicly (no anonymous posting).
- Teachers reply to doubts and offer clarifications.
- Students can also post specific academic problems and receive solutions.

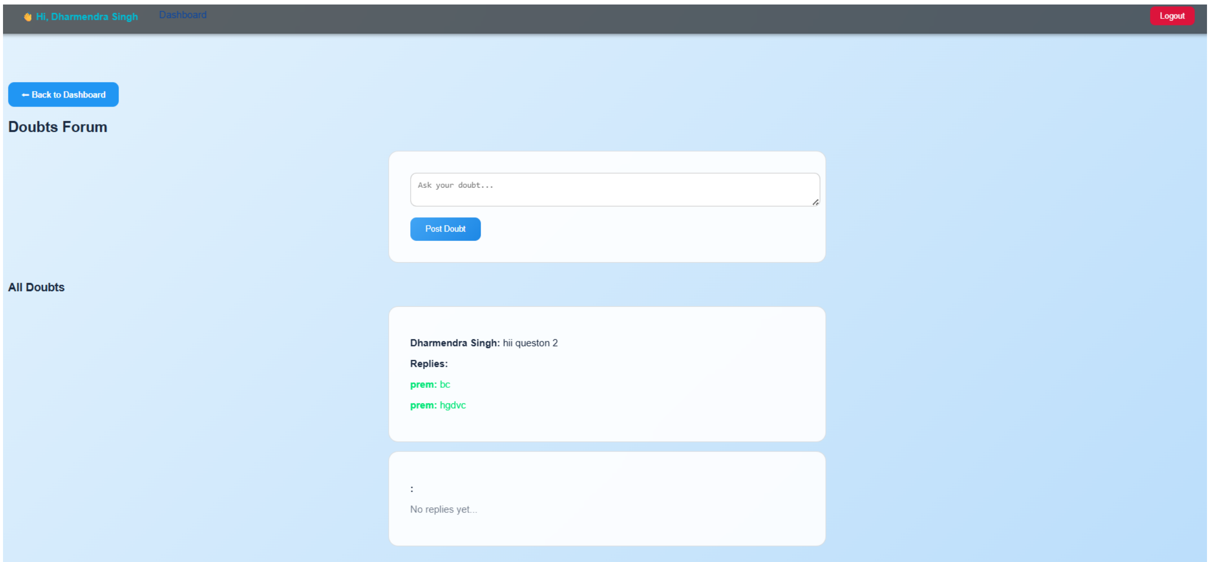

Figure 12: Doubt Submission Forum

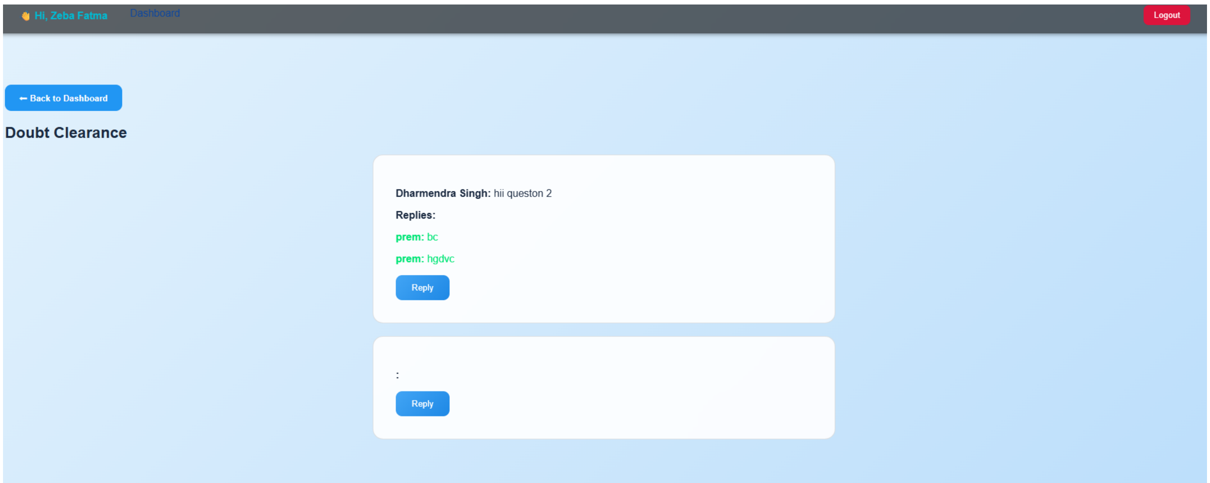

Figure 13: Teacher – Doubt Clearance Interface

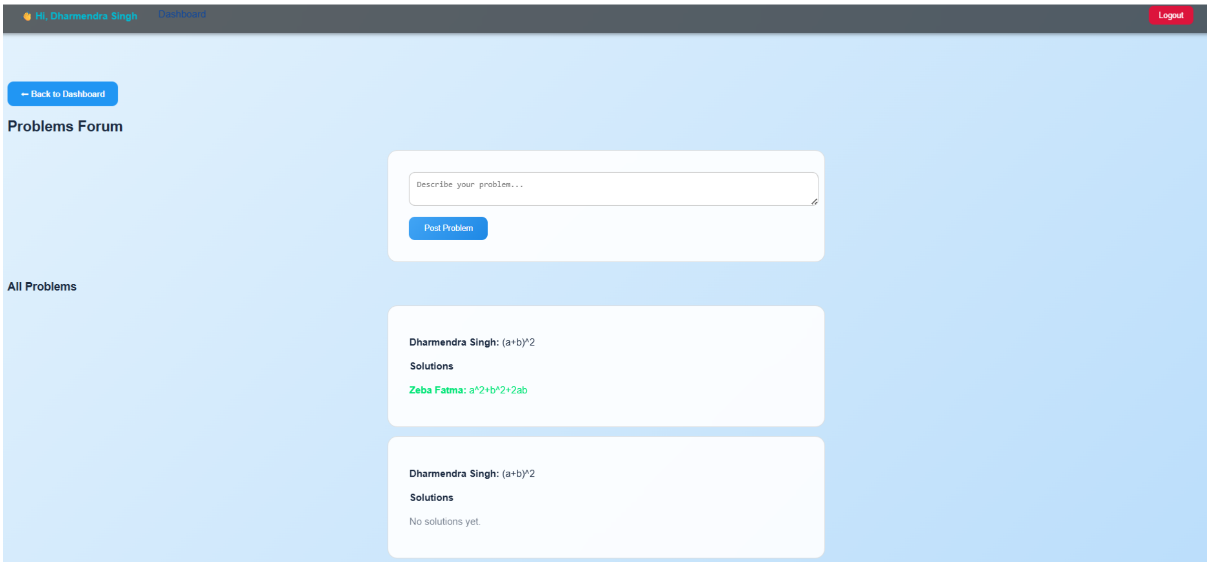

Figure 14: Problem Submission Forum

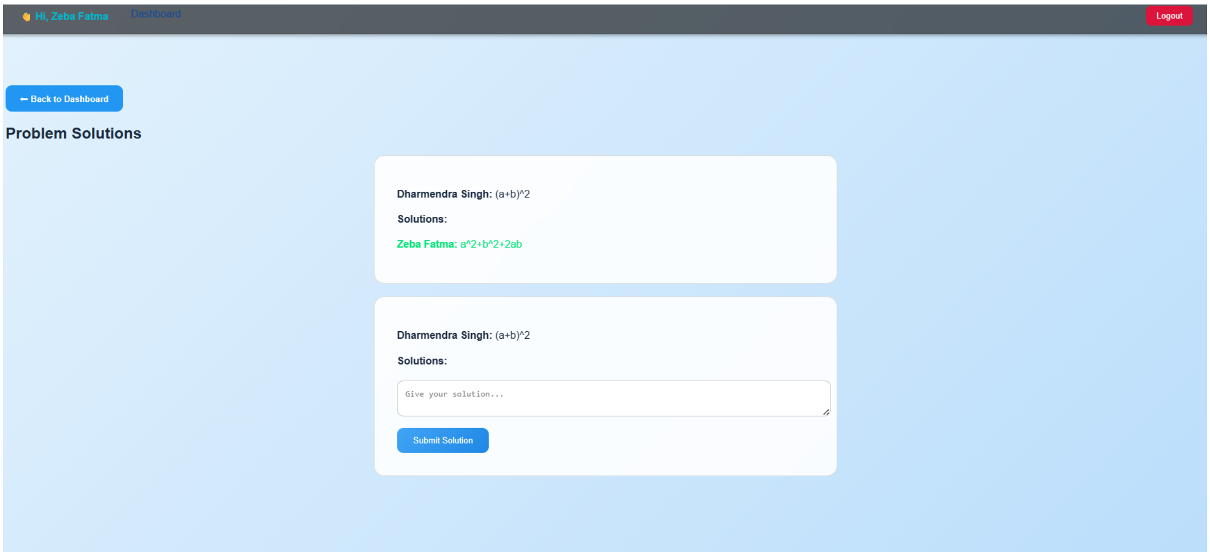

Figure 15: Problem Solutions Page

Student Performance Analytics

- Students can track performance across assignments and tests.
- Includes score trends, bar graphs, averages, and summaries.

Figure 16: Student – Performance Analytics Dashboard

Authentication and Dashboards

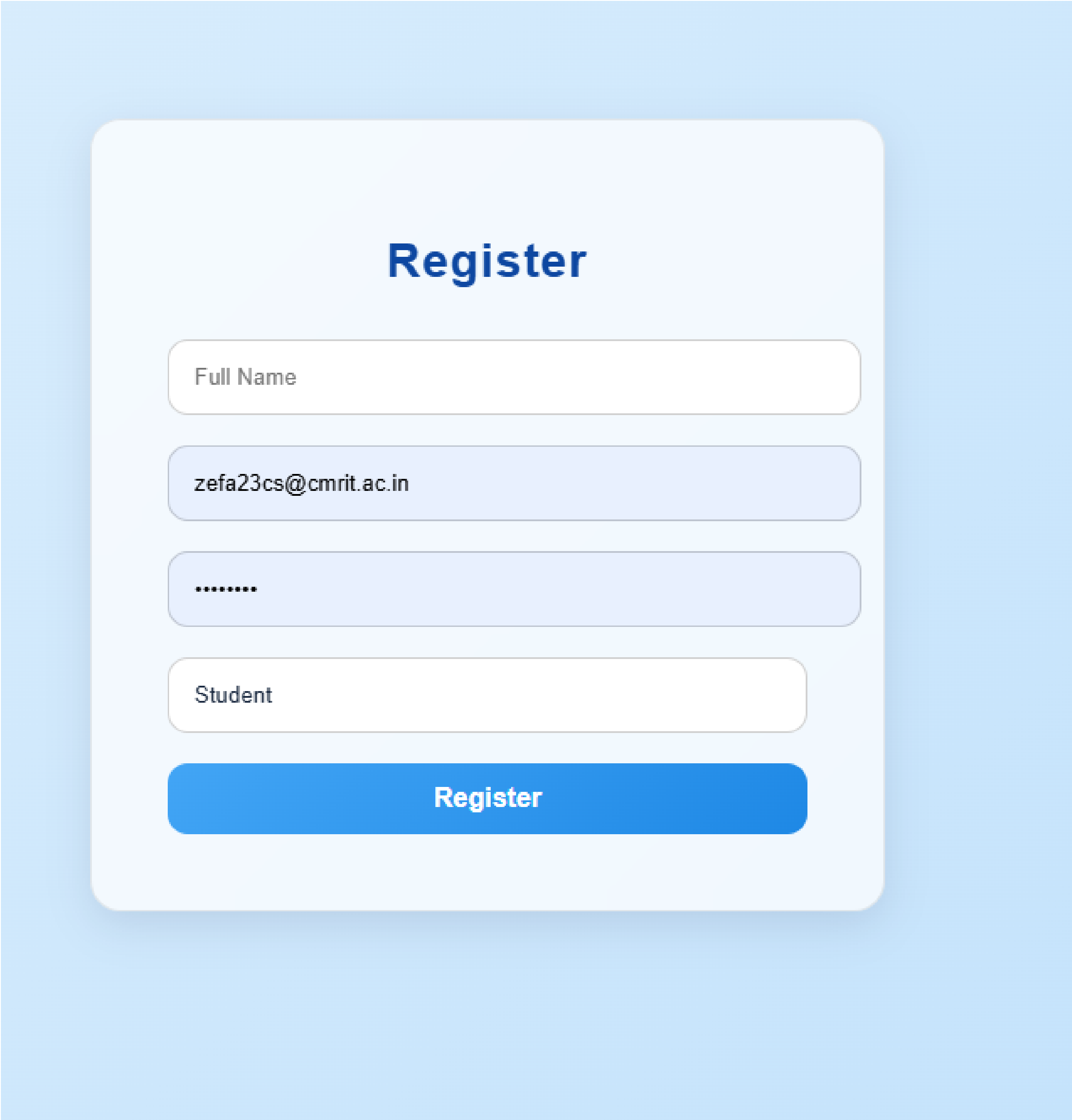

Figure 17: User Registration Page

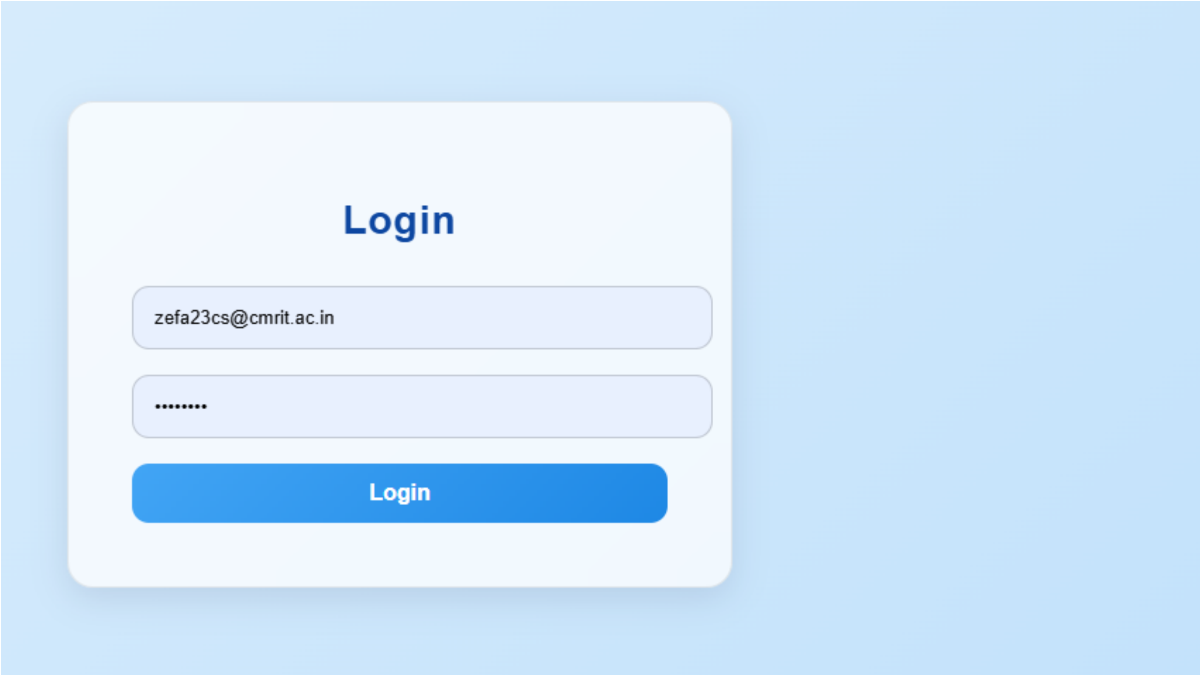

Figure 18: User Login Interface

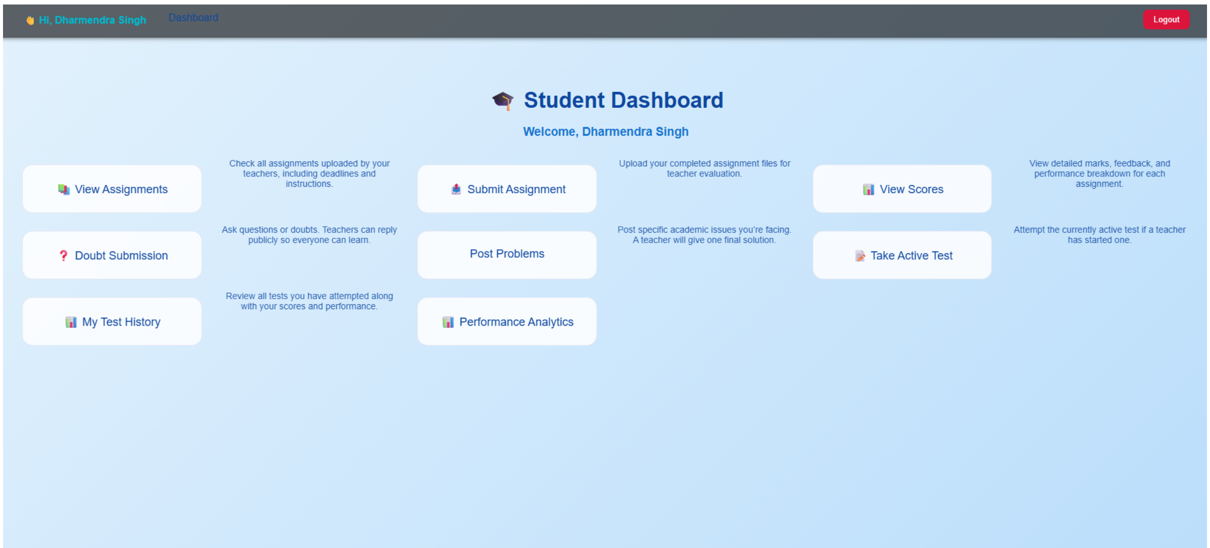

Figure 19: Student Dashboard

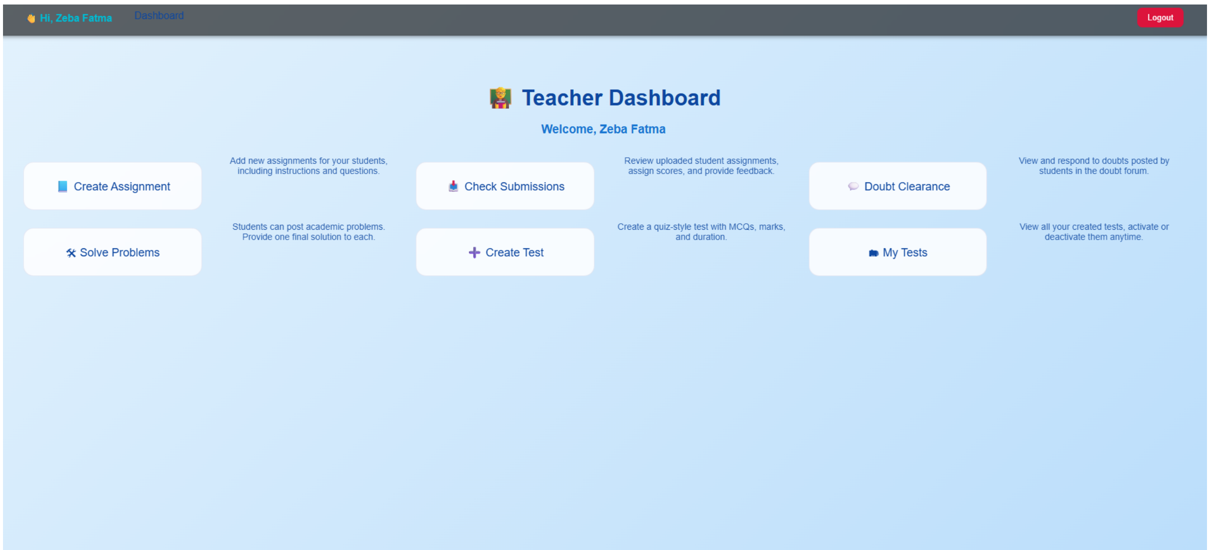

Figure 20: Teacher Dashboard

5. Conclusion

The Smart Assignment, Testing & Feedback Management System (SAFMS) provides a robust and modern solution for managing academic tasks in a digital environment. With the integration of MCQ-based online tests, real-time scoring, and a dynamic leaderboard, the system delivers a complete and efficient evaluation ecosystem.

By automating assignment handling, test conduction, and feedback processes, the system enhances productivity, transparency, and fairness for both teachers and students. Its modular and scalable architecture allows for easy expansion with features such as analytics, question banks, or AI-driven evaluation systems.

Overall, SAFMS acts as an effective bridge between educators and learners, offering a streamlined and technologically advanced approach to academic management.

6. References

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